

Customer Segmentation Analysis Report

Question 1: Report Objective

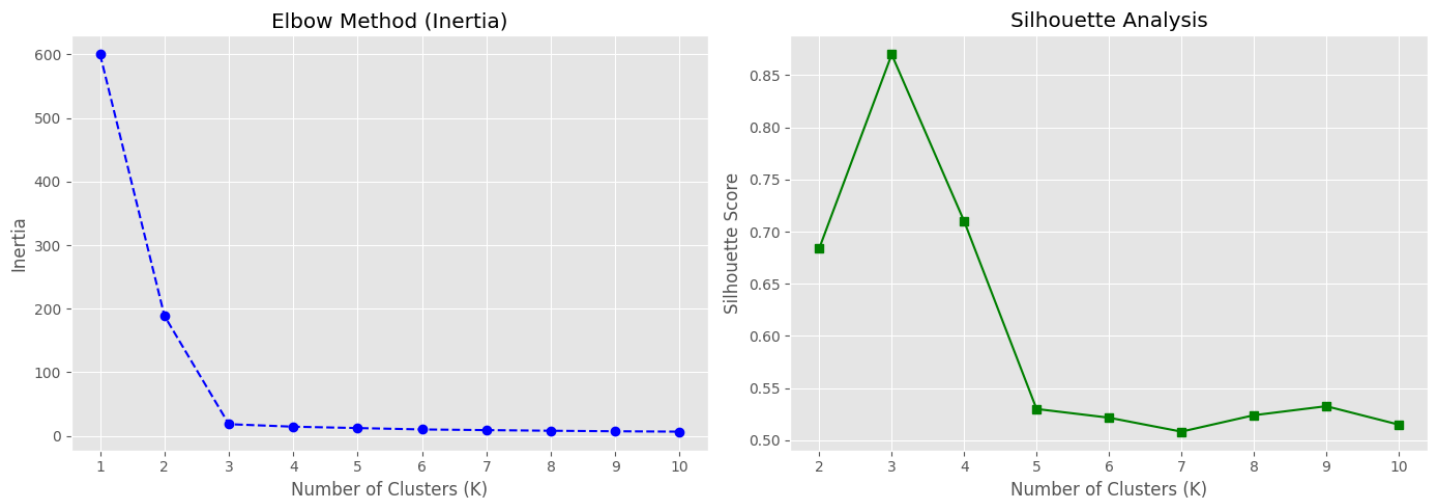
The objective of this report is to analyze followers’ engagement metrics using K-Means clustering to identify distinct user segments within the lifestyle brand's audience. This segmentation will provide actionable insights to help the marketing team create targeted campaigns that builds stronger community and brand loyalty.

Question 2: Context: The Impact of Feature Scale

Unscaled Data Summary			
	followers	posts_made	likes_received
count	200.000000	200.000000	200.000000
mean	1952.660000	472.615000	600.350000
std	2243.861781	438.343876	712.095766
min	0.000000	0.000000	21.000000
25%	94.750000	113.250000	49.750000
50%	648.500000	263.000000	162.000000
75%	4705.500000	1033.000000	1453.250000
max	6181.000000	1308.000000	2079.000000

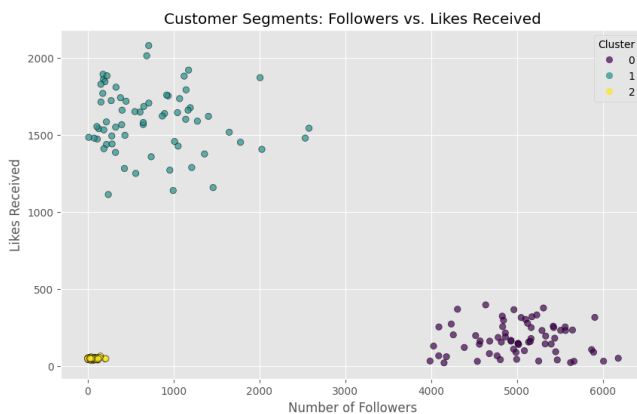
Looking at the unscaled data summary, the followers and likes_received features have significantly larger means and standard deviations compared to posts_made. If the data had not been scaled, the followers feature would have heavily and unfairly dominated the clustering algorithm. This occurs because K-Means assigns clusters based on calculating the geometric distance between data points. Consequently, the algorithm would treat large absolute differences in follower counts as massive gaps, while mathematically ignoring meaningful proportional differences in the much smaller posts_made metric. As a result, we would expect to see nonsensical clusters grouped almost entirely by follower size (e.g., "low," "medium," and "high" followers), completely ignoring the nuances of the users' actual posting and engagement behaviors.

Question 3: Selecting the Optimal Number of Clusters (K)



The Elbow Method was somewhat ambiguous for this dataset. While there is a general downward curve in inertia, the "elbow" where the drop-off stabilizes is not perfectly obvious, making it a subjective judgment call between $k=2$ and $k=3$. However, the Silhouette Score plot provided a much more decisive answer, showing a clear, definitive mathematical peak at $k=3$. Because the Silhouette Score actually evaluates how well-separated and cohesive the clusters are from one another, it was a more valuable metric for this particular analysis, confidently confirming $k=3$ as our optimal number of segments.

Question 4: Cluster Personas: Who Are These Users?



Cluster Summary (Mean values per cluster)			
	followers	posts_made	likes_received
cluster			
0	5017.253731	83.104478	175.373134
1	752.439394	1081.363636	1592.015152
2	70.373134	262.462687	48.462687

- **Cluster 0: "The Silent Audience"**: This group averages about 5017 followers, 83 posts and 175 likes. They represent users with a relatively high follower count but disproportionately low posting frequency and engagement, suggesting they are mostly passive content consumers.
- **Cluster 1: "The Highly Engaged Advocates"**: These users average roughly 752 followers, 1081 posts and 1592 likes. Despite having modest follower counts, they are the most active segment, creating a massive amount of content and generating the highest volume of engagement.
- **Cluster 2: "The Casual Everyday Users"**: Averaging around 70 followers, 262 posts and 48 likes, this segment represents standard personal accounts. They post moderately but only to a small circle, resulting in low overall reach and engagement.

Question 5: Actionable Business Recommendations

The client's primary goal is to increase brand loyalty and foster a stronger community feel, rather than focusing on immediate sales or massive follower growth. Based on this specific goal, Cluster 1 ("The Highly Engaged Advocates") represents the biggest opportunity. This segment is the key to building community because they already exhibit the desired behaviors: they are highly active, post frequently and drive significant engagement, making them genuine platform participants rather than passive scrollers. To leverage this group, the brand should launch a **User-Generated Content (UGC) Ambassador Program**. By inviting these highly engaged users to an exclusive insider community where they receive early access to products in exchange for honest reviews and community posts, the brand can directly reward their activity and solidify their loyalty.

Question 6: Model Limitations

One primary limitation of this analysis is the restricted set of features used. Incorporating additional data, such as user demographics (age, location) or sentiment analysis of their comments would have made the resulting segments even more insightful. Furthermore, a significant limitation of K-Means analysis is that it relies on data from a single point in time. User behavior is inherently dynamic, not static. A user currently in the "Casual" segment could have a post go viral tomorrow and immediately migrate to a different cluster, meaning this report only provides a temporary "snapshot". To provide the client with a more dynamic understanding of their audience over time the recommended next step is to implement time-series tracking. By continually collecting this same engagement data on a monthly basis, the agency can observe how users migrate between segments to measure the long-term impact of their marketing campaigns.